

INTERACTIVE TESTING & OPERATIONS MANUAL

The Complete System Walkthrough

A step-by-step companion guide designed for Daniel, partners, and stakeholders to test and explore CycleTrace alongside a live browser window.

Orientation for Daniel: This updated edition covers all recent platform enhancements including the self-service **Password Reset** workflow, **Stolen Bicycle Alerts** registry filter, **Store Owner Stock Handover with Verifiable Paper Trail Certificates**, and the new **Production Admin Passcode** (`cycletrace-admin`).

APPLICATION VERSION

CycleTrace 2.0 (Production)

PREPARED FOR

Daniel & Stakeholder Review

LIVE DEPLOYMENT

`cycletrace-v2.vercel.app`

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How to Use This Manual for Side-by-Side Testing

This manual is structured so that you can place this document side-by-side with an active browser window. Every subsection contains a highlighted **SIDE-BY-SIDE ACTION** box with exact URLs, button names, test input values, and expected UI updates.

Browser & Environment

Test on Google Chrome, Safari, or Edge pointing to <https://cycletrace-v2.vercel.app> (or local <http://localhost:3000>).

Live Cloud Persistence

CycleTrace connects directly to Appwrite Cloud and local registry caches. Actions persist in real time with zero mocked data.

System Architecture & Security

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CycleTrace is built with an immutable verification-first model designed for the realities of the South African cycling community. It bridges individual bicycle owners, independent bike shops, insurers, and law enforcement.

Technical Stack Overview

LAYER	TECHNOLOGY	FUNCTIONAL PURPOSE
Frontend	Next.js 15 (App Router), React 19, TypeScript	Server components, dynamic streaming layouts, and responsive client interfaces.
Authentication	Appwrite Auth, Session & Recovery API	Encrypted credentials, persistent sessions, and self-service account recovery tokens.
Database	Appwrite TablesDB & Local Synchronized Registry	Bicycle records, transfer records, organization applications, and verification queues.
Storage	Appwrite Storage Bucket (bikephotos)	Encrypted storage for bicycle photographs with strict access permissions.
Onboarding	OnboardJS React Tour Engine	Interactive product tour with local storage persistence and session synchronisation.
Payments	PayFast Integration	South African debit/credit card and Instant EFT subscription processing.
POPIA Protection	Data Minimisation Architecture	Public serial validation without ever exposing personal contact details or addresses.

Security & Privacy Principles

- **POPIA Compliance:** Public search results verify ownership authenticity and theft flags, but strictly withhold the owner's full name, phone number, and street address.
- **Role Separation:** Riders, Store Owners, Insurers, and Admins operate in isolated workspaces. Elevated roles require admin verification before privileged commercial tools unlock.
- **Cryptographic Handover:** Transfers utilize unique secure tokens tied to the designated recipient's email address to eliminate fraudulent serial claims.
- **Production Cleanliness:** Demo bypasses and direct admin links are hidden from user-facing screens to maintain strict security and production integrity.

Public Portal & Stolen Bicycle Alerts

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The public portal enables potential second-hand buyers, bike shops, and law enforcement to instantly verify a bike's ownership and theft status before money changes hands.

STEP 1 Landing Page & Live Serial Search

Navigate to [/](#). The homepage displays CycleTrace's mission, verification badges, and a prominent search bar.

SIDE-BY-SIDE ACTION:

Scroll to the search input under "Before you buy, check the trace". Enter a brand, model, or frame serial number (e.g. [Trek](#), [Specialized](#), or [WSBC](#)) and press Enter or click

[SEARCH REGISTRY](#) →

STEP 2 NEW: Stolen Bicycle Alerts Hotlist (</search?filter=stolen>)

Users and spotters can view all bicycles currently flagged as stolen across the country so stolen bikes can be spotted and reported.

SIDE-BY-SIDE ACTION:

1. Click the

 **STOLEN ALERTS**

link in the top navigation bar, or navigate to </search?filter=stolen>.

2. Notice the prominent red alert banner highlighting active theft cases.

3. Filter or search among reported stolen bikes by brand, color, or SAPS CAS case reference.

4. Click on any stolen bike card to inspect its distinguishing features, theft location, and recovery tip line.

STEP 3 Public Bike Record Verification ([/bikes/\[id\]](/bikes/[id]))

This page serves as proof of registry:

- **Protected Status:** Displays a green checkmark indicating the bike is legitimately registered and clear of theft reports.
- **Stolen Alert:** If flagged by the owner, a prominent red alert banner displays: *"This bike has been reported stolen. Do not purchase it."*
- **Frame Serial Verification:** Highlights the registered serial number so a buyer can inspect the physical bike frame on the spot.

Authentication & Self-Service Password Reset

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CycleTrace features unified authentication state management with automatic guest redirection and an official self-service password recovery system.

Authentication State Matrix

PAGE / LOCATION	SIGNED-OUT VISITOR	SIGNED-IN MEMBER
Homepage Navbar	Shows "Log in" and "Register a bike"	Shows "Dashboard" and "Sign out" button
Topbar Search / Nav	Shows "Log in" and "Get started"	Shows "Dashboard" and "Sign out" button
Login Route (<code>/login</code>)	Displays email, password, and "Forgot password?" link	Automatically redirects to <code>/dashboard</code>
Register Route (<code>/register</code>)	Displays 3-step member registration form	Automatically redirects to <code>/dashboard</code> (Unless in <code>?mode=bike</code>)
Password Recovery (<code>/forgot-password</code>)	Displays email recovery form with dispatch confirmation	Accessible for account management
Password Reset (<code>/reset-password</code>)	Validates recovery token and captures new password	Confirms update and directs user to log in

STEP 1 NEW: Requesting Password Recovery (`/forgot-password`)

SIDE-BY-SIDE ACTION:

- Navigate to `/login` and click **FORGOT PASSWORD?** next to the Password field.
- On `/forgot-password`, enter your registered account email (e.g. `you@example.com`).
- Click **SEND RECOVERY INSTRUCTIONS →**.
- Observe the green confirmation screen detailing that recovery instructions have been dispatched.

STEP 2 NEW: Executing Password Reset (`/reset-password`)

SIDE-BY-SIDE ACTION:

- Open the recovery URL received in your email (or test using `/reset-password?userId=...&secret=...`).
- Enter your new password (minimum 8 characters required).
- Confirm the password to ensure they match.
- Click **UPDATE PASSWORD →**.
- Observe the success confirmation card with the **BACK TO LOG IN** button.

Rider Dashboard & Onboarding Guide

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The Rider Dashboard (/dashboard) is the central command center for bicycle owners to oversee their registered fleet, inspect live records, and trigger recovery actions.

Interactive Onboarding Walkthrough (OnboardJS)

First-time users entering the dashboard are greeted with an interactive 3-step onboarding walkthrough guide:

STEP 1

Welcome to CycleTrace

Introduces the workspace where riders protect bikes and manage records.

STEP 2

Your Bikes Live Here

Points to the "My bikes" grid, tied to your secure Appwrite account ID.

STEP 3

Know Your Next Action

Shows how to register new bikes, transfer ownership, or flag theft.

SIDE-BY-SIDE ACTION:

1. Advance through the tour steps using the green

NEXT →

button.

2. Test the

BACK

button to return to a previous step.

3. Click

FINISH ✓

on the final step.

4. Notice the tour closes cleanly, and a floating

TOUR GUIDE

button appears in the bottom-right corner.

5. Click

TOUR GUIDE

to restart and replay the tour at any time.

Dashboard Key Metrics

- **Protected Bikes:** Total number of bikes registered under your Appwrite account currently in good standing.
- **Reported Stolen:** Real-time count of bikes actively flagged as stolen.
- **Account Status:** Confirms live active Appwrite session connectivity.

Adding Additional Bikes to Your Fleet

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When a logged-in rider wants to register another bicycle, they bypass account creation and jump directly to bicycle details.

STEP 1 Initiate Bike Registration (/register?mode=bike)

From the dashboard, click **Register another bike** or the sidebar/topbar **Add a bike** button.

SIDE-BY-SIDE ACTION:

Notice that Step 1 ("Account") is automatically marked ready. The screen starts directly on
01 BIKE DETAILS

STEP 2 Enter Bicycle Identifying Details

FIELD	EXAMPLE INPUT	VERIFICATION REQUIREMENT
Brand	Trek, Specialized, Cannondale, Giant	Exact manufacturer name.
Model	Domane SL 6, Stumpjumper, Synapse	Model and specification grade.
Year	2024	Model manufacturing year.
Colour	Matte Black / Hyper Yellow	Distinctive frame paint finish.
Serial Number	WTU12345678Z	Stamped on underside of bottom bracket.
Storage Location	Cape Town / Gardens	Primary residential city or suburb.
Photo Upload	Side profile image (< 10MB)	Clear picture showing frame geometry.

STEP 3 Review & Submit to Appwrite

Click **Continue** → to inspect the summary card. Confirm ownership accuracy checkbox, then click **Complete registration**.

EXPECTED OUTCOME:

A success confirmation screen displays your new unique
REGISTRATION ID
with links to view the public record or return to your dashboard.

Theft Reporting & Recovery Workflow

07

When a bicycle is stolen, every hour counts. CycleTrace allows immediate status escalation so any buyer or bike shop checking the serial number is warned.

STEP 1 Access Theft Reporting (/dashboard/report)

Navigate to `/dashboard/report` from your dashboard action menu.

SIDE-BY-SIDE ACTION:

1. Select the bicycle from the dropdown list of your owned bikes.
2. Enter the date of incident, last seen location, and police case number (SAPS CAS).
3. Add any distinguishing marks (e.g. scratched top tube, aftermarket pedals).
4. Click

FLAG AS STOLEN →

STEP 2 Verify Public Registry Escalation

CycleTrace updates the bike's record status in Appwrite from `protected` to `stolen` instantaneously.

SIDE-BY-SIDE ACTION:

Open an incognito or private browser tab. Go to `/search` and search for the serial number of the bike you just flagged. Observe the high-visibility red

REPORTED STOLEN

alert banner on its public page.

COMMUNITY PROTECTION MECHANISM:

The public page includes a direct reporting action for anyone spotting the bike to securely submit tips to `reports@cycletrace.co.za` without exposing the owner's personal details.

Secure Peer-to-Peer Ownership Transfer Protocol

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Selling or giving away a bicycle privately? CycleTrace enables verifiable cryptographic handovers, updating the provenance record and transferring management rights safely.

Two-Party Transfer Workflow

SENDER FLOW Initiating Transfer (/dashboard/transfer)

SIDE-BY-SIDE ACTION:

1. On `/dashboard`, locate the bike card and click **TRANSFER OWNERSHIP**.
2. Enter the buyer's email address (e.g. `buyer@example.com`).
3. Agree to release ownership rights and click **GENERATE TRANSFER LINK →**.
4. Copy the unique transfer URL containing the secure handover token.

BUYER FLOW Accepting Ownership (/transfer/[token])

SIDE-BY-SIDE ACTION:

1. Paste the transfer URL in a browser.
2. Review the bike details on the transfer reception page.
3. Enter the buyer email to verify authorization.
4. Click **ACCEPT OWNERSHIP →**.

POST-HANDOVER STATE:

The bike is immediately removed from the seller's active fleet and assigned to the buyer's account. The public provenance history logs the transfer timestamp.

Store Owner POS Stock & Commercial Paper Trail

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For bicycle retailers and bike shops, CycleTrace provides a specialized commercial workspace (`/store-owner`). Store owners register showroom stock, transfer ownership to customers upon sale, and issue verifiable digital & printable transfer certificates.

Store Owner Workspace Structure (`/store-owner`)

- **Store Profile & Verification Status:** Displays store trade name, physical address, business reg, and verified badge.
- **Inventory Metrics:** Live count of **Stock in store** versus bikes **Transferred to bikers**.
- **Shop Stock Inventory Tab:** Displays all bikes held in store stock with brand, model, serial, and a 1-click **Transfer to biker** action.
- **Customer Handover Paper Trail Log Tab:** Complete historical ledger of every customer handover, showing Date, Bike, Frame Serial, Customer Name & Email, Invoice Number, and direct link to the Certificate.

STEP 1 Registering Stock into the Showroom

SIDE-BY-SIDE ACTION:

1. On `/store-owner`, click **ADD STOCK BIKE**.
2. Enter the bicycle make, model, frame serial number, color, and photos.
3. Submit the bike. It appears under the **SHOP STOCK INVENTORY** tab ready for floor display.

STEP 2 Customer Handover & Transfer Form (`/store-owner/transfer`)

When a customer purchases a bicycle, the shop owner initiates a verified retail transfer that binds the sale to an invoice reference.

SIDE-BY-SIDE ACTION:

1. Click **TRANSFER TO BIKER** on any stock bike card.
2. Enter the Customer's **FULL NAME**, **EMAIL ADDRESS**, and **PHONE NUMBER**.
3. Enter the **STORE INVOICE / RECEIPT REF** (e.g. `INV-2026-0412`).
4. Set Handover Date and optional Warranty / Service notes.
5. Click **COMPLETE TRANSFER & GENERATE PAPER TRAIL →**.

STEP 3 Verifiable Transfer Certificate (`/store-owner/transfers/[id]/certificate`)

An official Certificate of Ownership Transfer is instantly generated. This serves as an audit-ready paper trail for the customer, store, and insurer.

SIDE-BY-SIDE ACTION:

1. Click

PRINT / DOWNLOAD CERTIFICATE

from the completion screen or from the Paper Trail Log.

2. Inspect the official verification seal, unique Certificate Serial Number (e.g. **CERT-CT-...**), frame serial, invoice reference, and dual signature sign-off lines.

3. Use **Ct r l+P / Cmd+P** to print a clean, high-resolution physical handover sheet for the customer.

Insurer Verification & Underwriting Portal

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The Insurer Workspace ([/insurer](#)) equips insurance underwriters and claims adjusters with tools to verify bicycle legitimacy, prevent policy duplication, and validate claims.

Insurer Capabilities

FEATURE	PURPOSE	OPERATIONAL VALUE
Policy Pre-Check	Verify serial authenticity before issuing cover	Eliminates phantom registrations and duplicate policies on the same frame.
Claims Verification	Cross-reference SAPS CAS and theft report status	Verifies whether a claimed bicycle was flagged stolen prior to the claim filing date.
Provenance History	Trace chain of custody from store sale to current owner	Provides clear evidence of legitimate ownership transfer and purchase invoices.
Fraud Hotlist Sync	Access real-time database of flagged suspicious serials	Reduces fraudulent claim payouts across South African underwriting firms.

ACTION Applying for Insurer Verification

SIDE-BY-SIDE ACTION:

1. Navigate to [/onboarding](#) and click **INSURER**

.

2. Enter your company name, FSP license number, underwriter contact, and business email.

3. Click

SUBMIT APPLICATION →

to queue for platform admin approval.

Platform Administration & Passcode Security

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The Platform Administration portal (`/admin`) allows platform governors to review pending store and insurer applications, approve credentials, and monitor ecosystem health.

Admin Passcode Security Model

ADMIN ACCESS PASSCODE:

The administration console is secured by an environment-level passcode. For Daniel and testing evaluators, the production passcode is configured as:

cycletrace-admin

Production Cleanliness & Security Enhancements

- **Concealed Admin References:** Public and guided experiences no longer display "View in Admin Dashboard" or "Verify this store in Admin" shortcuts. This guarantees real customers experience an authentic, production-ready interface.
- **No Demo Bypass:** The previous "Demo Quick Unlock" bypass button has been eliminated. Access requires explicit passcode authentication.
- **Real-Time Store Sync:** Store owner registrations persist instantly to the central registry. When a store owner applies, their application immediately appears in the Admin review queue.

ACTION Unlocking the Admin Dashboard

SIDE-BY-SIDE ACTION:

1. In your browser, navigate directly to `/admin` .
2. In the "Enter administrator passcode" prompt, input `cycletrace-admin` .
3. Click
UNLOCK ADMINISTRATOR CONSOLE →
.
4. Inspect the
PENDING APPLICATIONS
queue. Click
VERIFY STORE
on any newly registered bike shop.
5. Switch back to the store owner window: the shop is now immediately verified and unlocked!

Account Preferences & Contact Security

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Members manage their personal contact information, regional alert city, Appwrite session security, and subscription billing via `/settings`.

Account Preferences (`/settings`)

- **Full Name:** Updates user display name across ownership records, certificates, and transfers.
- **Email Address:** Read-only display of primary Appwrite account identity.
- **Contact Phone:** Stored securely for recovery notifications and customer handover certificates.
- **Home City:** Primary locality for regional recovery network alerts and stolen alerts matching.
- **Security:** Direct visibility of active Appwrite Auth session encryption and account status.

ACTION Updating Phone & Home City

SIDE-BY-SIDE ACTION:

1. Navigate to `/settings` while logged in.
2. Enter your cell phone number (e.g. `+27 82 123 4567`) and your Home City (e.g. `Cape Town` or `Johannesburg`).
3. Click

SAVE PREFERENCES →

4. Observe the confirmation badge confirming your contact preferences have been synced with Appwrite.

Billing & Subscription (`/settings/billing`)

CycleTrace integrates with South Africa's leading payment gateway, **PayFast**, to manage monthly bike registry subscriptions (R24.99 / month). Active Appwrite records are maintained with live subscription status tracking.

Daniel's End-to-End Testing Checklist

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This verification checklist is prepared specifically for Daniel and the evaluation team to test each core feature of the live system.

✓	FEATURE AREA	VERIFICATION STEPS FOR DANIEL	EXPECTED RESULT
[]	1. Public Search	Go to <code>/search</code> , query a bike serial number or brand.	Matching bikes appear with photo and protected badge.
[]	2. Stolen Alerts Filter	Click  Stolen alerts in topbar or visit <code>/search?filter=stolen</code> .	Displays active stolen hotlist with red alerts and tip submission links.
[]	3. Password Reset Request	From <code>/login</code> , click Forgot password? . Submit email on <code>/forgot-password</code> .	Green confirmation card confirms recovery instructions sent.
[]	4. Password Reset Execution	Open <code>/reset-password?userId=...&secret=...</code> , enter 8+ char password.	Password successfully updated; redirects to login.
[]	5. Auth Smart Redirect	While logged in, attempt to navigate to <code>/login</code> or <code>/register</code> .	Instantly redirects to <code>/dashboard</code> without flashing forms.
[]	6. Onboarding Tour	Open <code>/dashboard</code> . Advance through the 3-step tour and replay.	Tour guides user smoothly; "Tour guide" button persists.
[]	7. Fleet Registration	From dashboard, click Register another bike .	Skips account step; captures frame specs, serial, and photo.
[]	8. Theft Flagging	Navigate to <code>/dashboard/report</code> and flag an owned bike as stolen.	Status updates to stolen; public search displays high-visibility alert.
[]	9. Peer Transfer	Generate peer transfer token on <code>/dashboard/transfer</code> .	Claimable on <code>/transfer/[token]</code> ; ownership reassigns.
[]	10. Store Stock Registration	In <code>/store-owner</code> , add a showroom bike to inventory.	Bike appears in Shop Stock Inventory tab.
[]	11. Store Stock Handover	Click Transfer to biker on stock bike. Fill customer details & invoice ref.	Updates stock to transferred; logs paper trail in store ledger.
[]	12. Transfer Certificate	View certificate on <code>/store-owner/transfers/[id]/certificate</code> .	Displays official seal, certificate number, invoice ref & signatures.
[]	13. Admin Passcode Console	Navigate to <code>/admin</code> . Enter <code>cycletrace-admin</code> .	Console unlocks; allows 1-click verification of store applications.
[]	14. Contact Settings	Update Phone Number and Home City on <code>/settings</code> and save.	Preferences save cleanly and persist in Appwrite user profile.

Frequently Asked Questions & Support

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Common Questions

Where do I find my bike's serial number?

On 95% of bicycles, the serial number is stamped into the metal on the underside of the bottom bracket shell (beneath the pedals). It can also sometimes be found on the rear dropout or the head tube.

What is the difference between Peer Transfer and Store Stock Transfer?

Peer transfers (</dashboard/transfer>) are for private individuals selling used bikes. Store Stock Transfers (</store-owner/transfer>) are point-of-sale commercial handovers executed by verified retailers, capturing store till slip/invoice references and generating printable Certificates of Ownership Transfer with official serial numbers.

Can someone steal my bike by claiming my serial number?

No. Once a serial number is registered in CycleTrace, any attempt to register the same serial number triggers an automatic **409 Conflict** guard. Ownership can only be updated through an authorized cryptographic handover link from the existing registered owner.

How does CycleTrace protect owner privacy under POPIA?

In accordance with the South African Protection of Personal Information Act (POPIA), public search results only disclose bicycle technical specifications and theft verification status. Your full name, residential address, and contact numbers are strictly confidential.

Technical Support & Enquiries

For technical inquiries, partner onboarding, or pilot deployment support, please contact the CycleTrace team:

- **General Enquiries:** hello@cycletrace.co.za
- **Theft Recovery & Law Enforcement:** reports@cycletrace.co.za
- **Privacy & Compliance:** privacy@cycletrace.co.za
- **Official Website:** <https://cycletrace.co.za>
- **Live System Deployment:** <https://cycletrace-v2.vercel.app>